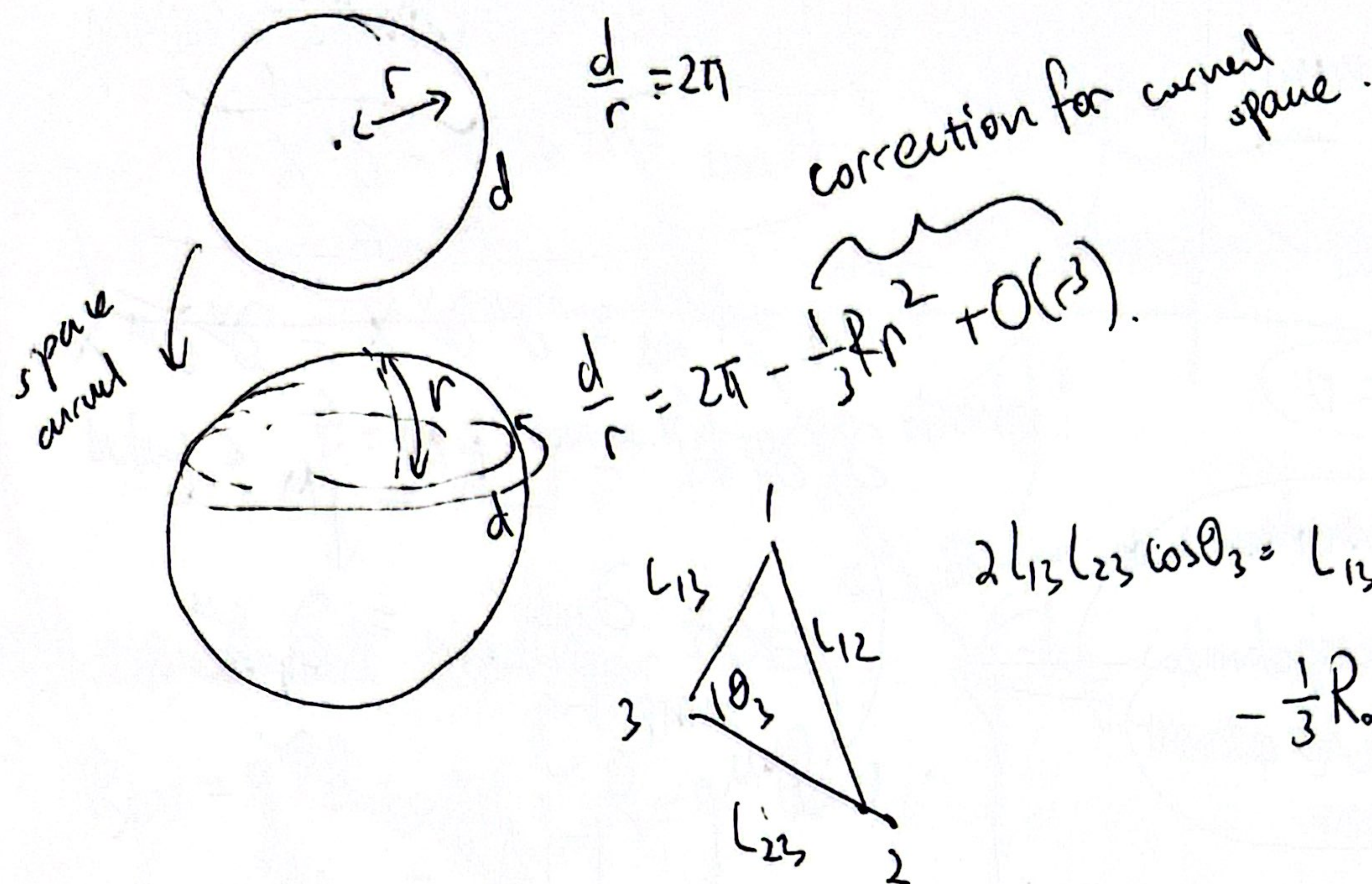


$$R^\mu_{\alpha\beta\gamma} = \Gamma^\mu_{\alpha\gamma,\beta} - \Gamma^\mu_{\alpha\beta,\gamma}$$



$$2L_{13}L_{23}\cos\theta_3 = L_{13}^2 + L_{23}^2 - L_{12}^2$$

$$-\frac{1}{3}R_{\alpha\beta\gamma\delta}(\Delta x_{13})^\alpha(\Delta x_{23})^\beta(\Delta x_{13})^\gamma(\Delta x_{23})^\delta + O(\Delta x^5).$$

$\frac{d^2\epsilon}{d\lambda^2}$ ← didn't try
get that pt.

Draw Connections btwn Maxwell's Laws.

$$\underbrace{\nabla_\nu F^{\mu\nu}}_{\text{fields.}} = \int \underbrace{\rho}_{\text{sources}} dV$$

$$\underbrace{\hspace{10em}}_{\text{fields}} = \underbrace{\hspace{10em}}_{\text{source}} T_{\mu\nu} \rightarrow \text{conserved.}$$

energy
momentum
tensor

$$\nabla^\mu T_{\mu\nu} = 0, \quad T_{\mu\nu} = T_{\nu\mu}.$$

better be

some scalar

$$(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R) + \Lambda g_{\mu\nu} =$$

Λ and

what is this constant?
 $8\pi G_N$

source

Newton's Constant

what are the symmetric, conserved tensors we know?
conserved & symmetric.

Why $\vec{F} = m\vec{a}$ true? $\rightarrow$ It works!

Λ does not really matter.

10 coupled non-linear PDEs.

$$R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu}, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

the cosmological constant needs to be measured experimentally.

Has dimensions of energy density

$$\Lambda \sim 5.96 \times 10^{-27} \text{ kg/m}^3.$$

$$\sim 4 \text{ protons/m}^3.$$

when divided by G_N .

So small it's irrelevant

$$\left[R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G_N T_{\mu\nu} \right] \rightarrow$$

How can you see apples fall (ordinary $\vec{F} = m\vec{a}$) from this?

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G_N \left(T_{\mu\nu} - \underbrace{\frac{1}{8\pi G_N} g_{\mu\nu}} \right)$$

just looking at this you can't
 see that this contributes
 to the expansion of the universe.

"Weak Field" limit.

$$g_{\mu\nu} = \underbrace{\eta_{\mu\nu}}_{\text{Minkowski}} + h_{\mu\nu} \quad \begin{array}{l} \text{small} \\ \text{correction} \end{array} \quad |h_{\mu\nu}| \ll 1$$

every component.

$$\eta = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}$$

All computations to linear order in h . (Neglect $O(h^2)$).

$$(1) \quad g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$$

↑
 raised with η (not g).
 $h^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} h_{\alpha\beta}$

$$\begin{aligned} \delta_\mu^\nu &= g_{\mu\sigma} g^{\sigma\nu} \\ &= (\eta_{\mu\sigma} + h_{\mu\sigma}) (\eta^{\sigma\nu} - h^{\sigma\nu}) \\ &= \delta_\mu^\nu - h_\mu^\nu + h_\mu^\nu + O(h^2) \end{aligned}$$

$$(2) \quad \text{L.H.S.} \quad R_{\alpha\beta\gamma\delta} = T - T + T T - T T$$

$$T = \frac{1}{2} g^{-1} (g_{-} + g_{-} - g_{-})$$

1st deriv $T \sim h$.

For small h , $T T$ terms can be
 neglected.

$$R_{\alpha\beta\gamma\delta} = \frac{1}{2} (g_{\alpha\delta, \beta\gamma} - g_{\beta\delta, \alpha\gamma} + g_{\beta\gamma, \alpha\delta} - g_{\alpha\gamma, \beta\delta})$$

$$R_{\alpha\beta\gamma\delta} |_{g=\eta+h} = \frac{1}{2} (h_{\alpha\delta, \beta\gamma} - h_{\beta\delta, \alpha\gamma} + h_{\beta\gamma, \alpha\delta} - g_{\alpha\gamma, \beta\delta}) \quad \begin{array}{l} \text{trace} \\ \downarrow \\ h = h^\alpha{}_\alpha \end{array}$$

$$\cancel{R_{\mu\nu}} = R_{\beta\delta} = \eta^{\alpha\gamma} R_{\alpha\beta\gamma\delta} = \frac{1}{2} (h^\gamma{}_{\delta, \beta\gamma} - h_{\beta\delta, \gamma}{}^\gamma + h^\gamma{}_{\beta, \gamma\delta} - h^\alpha{}_{\alpha, \beta\delta})$$

$$R = \eta^{\beta\delta} R_{\beta\delta} = \frac{1}{2} (\underbrace{h^{\gamma\delta}{}_{,\gamma\delta}} - \underbrace{h^{\delta,\gamma}{}_{,\gamma}} + \underbrace{h^{\gamma\delta}{}_{,\gamma\delta}} - \underbrace{h^{\delta,\gamma}{}_{,\gamma}})$$

$$= \frac{1}{2} (h^{\gamma\delta}{}_{,\gamma\delta} - \square h)$$

$$R_{\beta\delta} - \frac{1}{2} \eta_{\beta\delta} R = \frac{1}{2} (h^\gamma{}_{\delta, \beta\gamma} + h^\gamma{}_{\beta, \gamma\delta} - \square h_{\beta\delta} - h_{\beta\delta, \gamma}{}^\gamma)$$

$$+ \eta_{\beta\delta} \square h - \eta_{\beta\delta} (\underbrace{h^{\mu\nu}{}_{,\mu\nu}}) = 8\pi G_N$$

$h_{\mu\nu}$ is like the potential $\rightarrow$ (potential defined up to a constant).

$$A^\mu = \begin{pmatrix} \phi \\ \vec{A} \end{pmatrix} \rightarrow \vec{A}$$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

$$A^\mu \rightarrow A^\mu - \partial^\mu \phi$$

(defined up to a certain function??)

$$\frac{1}{2} \square h$$

$$\square^{\alpha}{}_{\alpha} = \frac{\partial}{\partial x^\alpha} \frac{\partial}{\partial x^\alpha} \eta^{\alpha\beta}$$

$$= -\frac{\partial^2}{\partial t^2} + \vec{\nabla}^2$$

$$= \square$$

$\uparrow$
d'Alembertian

$$x^\mu = x^{\mu'} - \underbrace{\xi^{\mu'}}_{\substack{y \\ \text{small}}}(x')$$

$$g_{\mu\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} g_{\mu\nu}$$

new/old coordinates.

$$\eta_{\mu\nu'} + h_{\mu\nu'}^{(\text{new})} = \left(\delta_{\mu'}^\mu - \xi_{\mu'}^\mu \right) \left(\delta_{\nu'}^\nu - \xi_{\nu'}^\nu \right) \left(\eta_{\mu\nu} + h_{\mu\nu}^{(\text{old})} \right)$$

$$h_{\mu\nu}^{(\text{new})} = h_{\mu\nu}^{(\text{old})} - \xi_{\mu\nu} - \xi_{\nu\mu}$$

choose $\xi : \left(h^{\alpha\beta} - \frac{1}{2} \eta^{\alpha\beta} h \right)_{,\beta} = 0$

↑
Lorentz Gauge

$$h^{\alpha\beta}_{,\beta} = \frac{1}{2} \eta^{\alpha\beta} h_{,\beta}$$

$$h^{\alpha\beta}_{,\alpha\beta} = \frac{1}{2} \eta^{\alpha\beta} \underbrace{h_{,\alpha\beta}}_{\text{why}} = \frac{1}{2} \square h$$

$$\square \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) = (-16\pi G_N) T_{\mu\nu}$$

how did you get this?

empty space $T_{\mu\nu} \approx 0$.

$$\eta^{\mu\nu} \left[\square \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) \right] = 0$$



$$\square \left(h - \frac{4}{2} h \right) = 0$$

$$\square h_{\mu\nu} = 0$$

$$\square h = 0$$

$$\square A_\mu = 0$$

these must be waves
that travel at c .

$\Rightarrow$ this
a ruler?

when we are contracting $R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu}$
what are we really doing? (physically)?

2nd derivatives = curvature.

why do we need a metric?